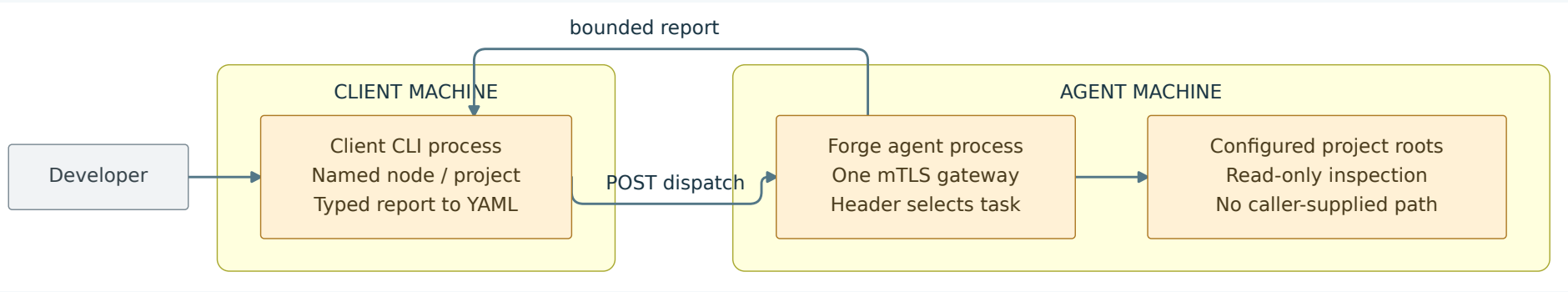
Remote Forge IDE proposal

Next milestone: run a configured node agent and reuse local project inspection remotely. Keep one front door and typed contracts. This is a proposal, not a deployed remote IDE.

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What exists today

A local IDE inspector and an authenticated loopback **node.info** proof. The current terminal, jobs and project inspection remain local.

What this milestone adds

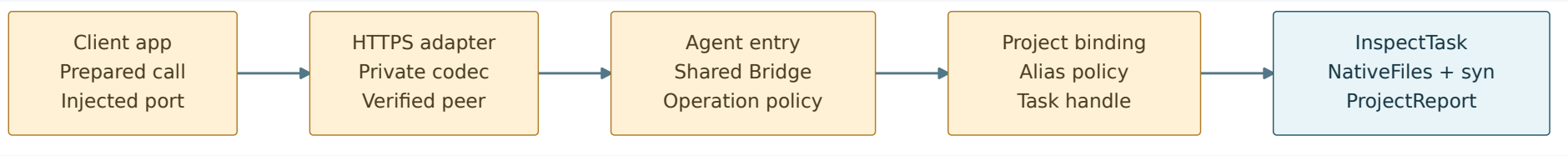
Named nodes, an agent lifecycle and an allowlisted project alias. The agent returns a typed report; the caller renders YAML.

Current HTTPS boundary Current IDE scope

Keep capabilities behind one front door

Thin adapters project input into a canonical envelope. The proposed agent app owns client and server composition; capabilities do not import HTTP, certificates or the agent application.

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Proposed package placement

NEW APPLICATION

forge-agent

Client and server composition
Local CLI bridge binding
config / catalog / policy
dispatch / lifecycle

PROPOSED ADAPTER EXTENSION

forge-remote-https

Private IDE codec and typed gateway seam
Implements the capability port
Never imports forge-agent

EXISTING CAPABILITY

forge-ide

task / filesystem / syntax / model
Proposed optional remote contract
Existing CLI remains local

Cargo direction Application imports adapters and capability contracts. The capability never imports the application or transport.

Endpoint is not task routing

A configured node name selects the network endpoint and expected identity. Canonical **Header** destination and version select a registered operation.

Retain the existing capability

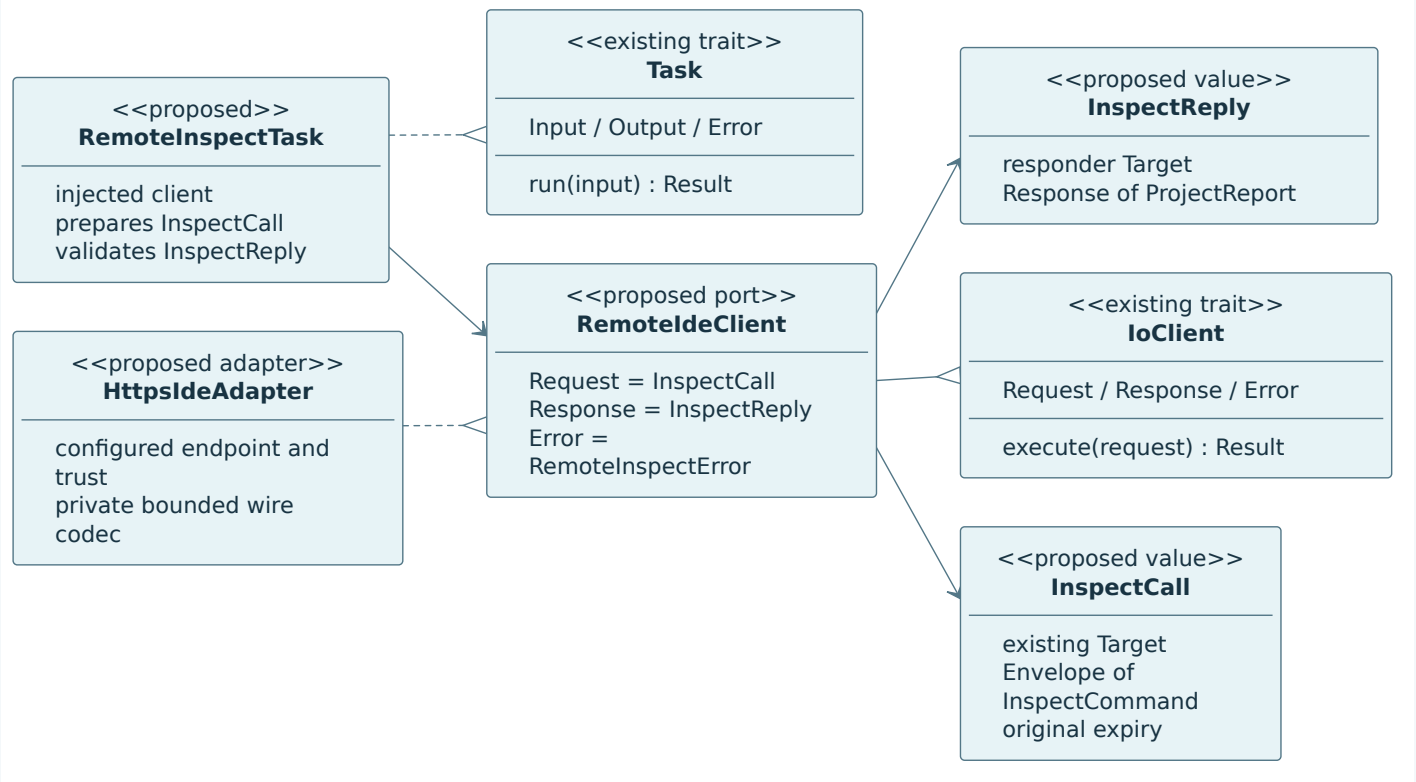
Reuse **forge-ide** model, task, filesystem and syntax modules for trusted projects. Reports are not secret-redacted; transport and YAML stay at the edges.

Remote typed conversation IDE module boundaries

Keep contracts independent of transport

UML style names summarize responsibilities rather than reproduce Rust signatures. The generic execution and I/O contracts already exist; remote IDE contracts are proposed.

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Actual contracts

Task owns Input, Output and Error; **IoClient** owns Request, Response and Error. Factories construct; providers control acquisition and leases.

Proposed composition boundary

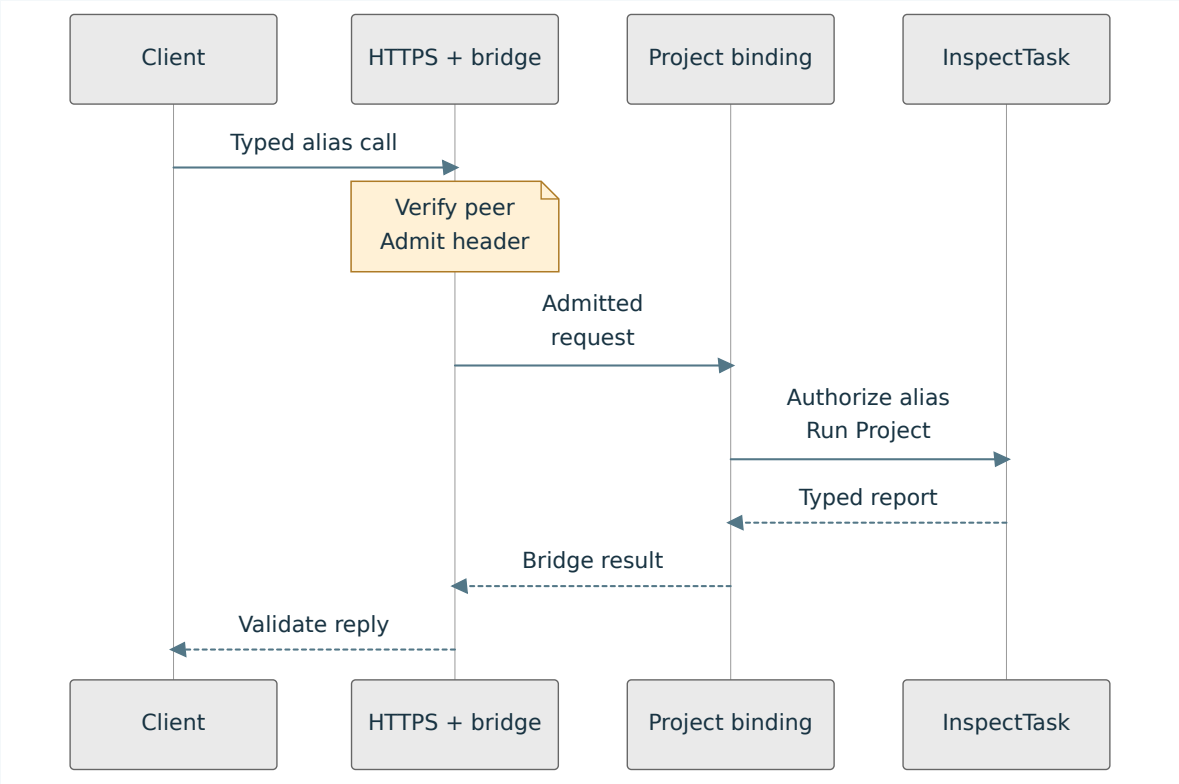
A proposed remote inspection client carries a typed request and report. The current **Destination** is node-only; it is not already an IDE dispatcher.

Task contract I/O client contract Current UML gallery

Admit identity before inspecting a project

Keep POST /forge/v1/dispatch and header selected routing. Verified TLS identity is not an HTTP claim, and authentication never replaces operation and project authorization.

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Reject before capability work

Check target, version, typed payload, expiry and policy. Resolve a permitted project alias to a configured root; do not accept an arbitrary remote path.

Validate the return path

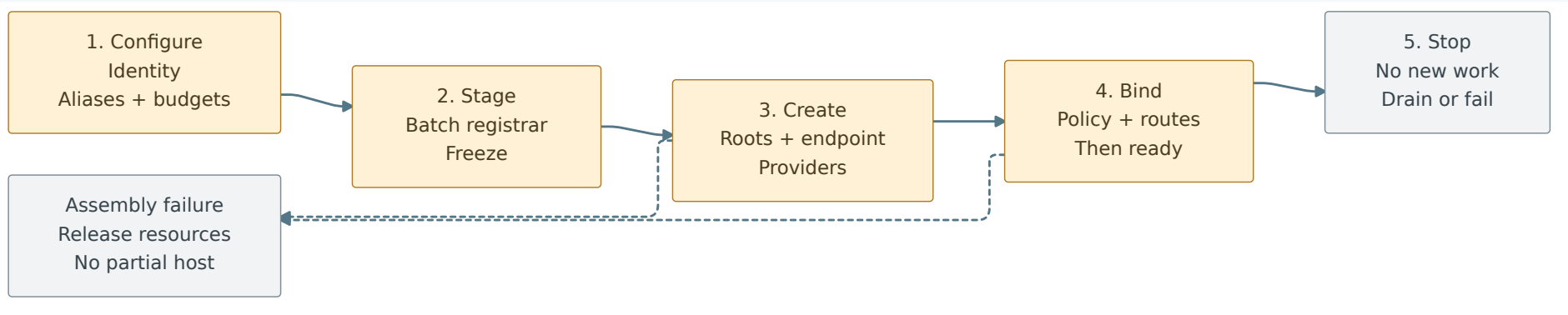
Bound the report and check responder and correlation metadata. A timeout or lost reply can follow execution; do not silently retry or claim cancellation.

Header and TLS rules Authorization and reply checks

Compose once before serving requests

Preserve the typed registrar and factory phases. The proposed agent owns explicit startup, readiness, request serving and shutdown instead of turning a loopback example into a daemon.

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Configuration before effects

Stage one proposed batch group for runtime project aliases. Reject conflicts, freeze, then invoke factories. Inject providers and tasks; publish readiness only after successful assembly.

Production setup is deliberate

Host certificates, peer mappings, project roots and access policy need explicit setup. No automatic trust store, firewall or fleet provisioning is part of this proposal.

Existing bootstrap sequence Current adapter construction

Ship a read only milestone first

Finish a small remote observation path before adding jobs or mutations. Every stage needs evidence of its failure behavior as well as its successful result.

01

Agent lifecycle

Config, readiness and shutdown

02

Remote inspection

Named node and permitted project

03

Bounded remote jobs

Status, ownership and cancellation

04

Mutations later

Explicit contracts and authorization

NEXT MILESTONE

LATER NOT INCLUDED

Acceptance evidence required before calling it done

Identity and access	Wrong peer, wrong target and denied project fail before inspection.
Network and protocol	Unreachable node, malformed input, unknown route and incompatible version fail clearly.
Bounded work and output	Entry, file, parser and response limits remain explicit; oversized output is rejected.
Timeout and lifecycle	Prove start and shutdown behavior; report execution uncertainty after a lost reply.
Typed result and features	Preserve report completeness; render YAML only at the edge; omit unselected transport dependencies.

Not in this milestone Remote shell or arbitrary paths, a full editor or compiler service, automatic device provisioning, complete GET PUT PATCH DELETE semantics, and Bitwave. Bitwave remains an optional future codec or carrier, not a prerequisite.

Read the full proposal Open the design skill